

Descriptive Statistics and Introduction to Probability
Problem sheet N°3

Exercise 1. We carried out the adjustment of a points of a couple of statistical variables (X, Y) . The equations obtained are the following:

For the regression line $D_{Y/X} : y = x + 30$,

For the regression line $D_{X/Y} : x = \frac{1}{4}y + 60$.

1. Deduce the value of the linear correlation coefficient.
2. Calculate the arithmetic means of the variables X and Y .
3. Calculate the covariance between X and Y , given that the variance of Y is equal to 40.

Exercise 2. The braking distance of a truck driving on a dry road with the respect to its speed is given by the following table

Speed (km/h)	30	40	50	60	70	80	90	100	110	120
Distance (m)	11	15	25	35	47	62	78	95	115	145

1. Calculate the means $\bar{X}$ and $\bar{Y}$ and the standard deviations σ_X and σ_Y of the variables X and Y as well as their correlation coefficient.
2. Draw the scatter plot. What kind of connection is there between X and Y ? Justify.
3. Determine the regression line of Y on X . Plot this line on the same graph. Show the center of gravity $(\bar{X}, \bar{Y})$.
4. Does the regression line fit the data closely? Measure it.
5. Use this equation to estimate the braking distance if the truck is traveling at 130 km/h.

We give $\sum_{i=1}^{10} x_i = 750$, $\sum_{i=1}^{10} y_i = 628$, $\sum_{i=1}^{10} x_i y_i = 59100$, $\sum_{i=1}^{10} x_i^2 = 64500$, $\sum_{i=1}^{10} y_i^2 = 57608$.

Exercise 3. Let P be the percentage of executives of a company whose net monthly salary is greater than the value R (in $10^4 DA$)

R	8	10	15	20	35	50
P	80.3	55.8	28.1	15.2	5.1	2.5

1. Plot the scatter plot. What kind of adjustment can we suggest?
2. Let $x_i = \log(R_i)$ and $y_i = \log(P_i)$. Draw the scatter plot (x_i, y_i) .
3. Calculate the linear correlation coefficient $\rho(X, Y)$. Comment.
4. Give the equation of the regression line $y = ax + b$ and the equation of the curve $P = f(R)$.
5. Calculate P if $R = 77060$.

Exercise 4. A sample of the evolution of electricity consumption in mega Wh between 1975 and 2003 is given by

X (Year)	1975	1979	1983	1987	1991	1995	1999	2003
Y (consumption)	60	82	112	146	194	246	330	414

1. Represent the scatter plot corresponding to these observations.
2. We set $u_i = \frac{x_i - 1989}{9.798}$ and $v_i = \log(y_i)$. Calculate the $(u_i, v_i)'$ s, represent the results using a table then in a scatter plot of the corresponding points. In view of the result obtained, what do you think is the purpose of the previous transformations?
3. We denote by (U, V) the pair whose observations are the $(u_i, v_i)'$ s. Calculate $\bar{U}, \bar{V}, \sigma_U^2, \sigma_V^2, \text{cov}(U, V)$ and $\rho(U, V)$.
4. Write the linear regression equation of V on U then measure its goodness of fit.
5. Deduce a relation between Y and X .
6. Assuming the evolution of electricity consumption followed during the years after the same rhythm, give an estimate of the consumption in the year 2007.